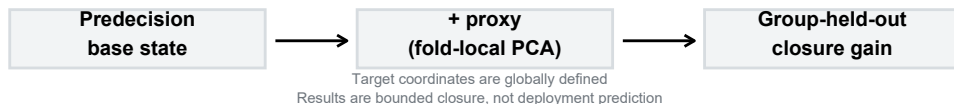


Fold-local proxy reduction preserves the hidden-state closure advantage

a



b

Proxy-specific closure gains across checkpoints

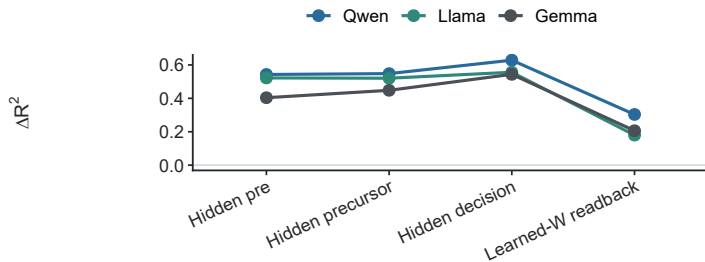
Each cell is augmented minus base performance



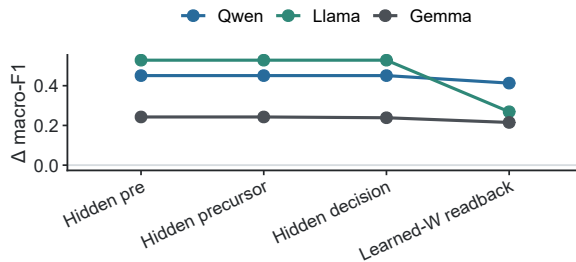
	Qwen				Llama				Gemma			
Random K	-0.02	-0.02	-0.02	-0.03	-0.01	-0.00	0.02	-0.00	-0.01	-0.00	-0.02	-0.00
Gaussian readback	0.30	0.23	0.34	0.30	0.16	0.04	0.29	0.05	0.17	0.02	0.15	0.01
Row-permuted-W	0.30	0.30	0.41	0.39	0.18	0.03	0.27	0.04	0.21	0.02	0.22	0.02
Learned-W readback	0.30	0.30	0.41	0.39	0.18	0.03	0.27	0.04	0.21	0.02	0.22	0.02
H-shape	0.25	0.65	0.28	0.73	0.07	0.24	0.15	0.30	0.09	0.10	0.06	0.08
Direct hidden flow	0.45	0.37	0.44	0.47	0.34	0.07	0.51	0.10	0.28	0.04	0.23	0.03
Hidden decision	0.63	0.42	0.45	0.55	0.56	-0.14	0.53	-0.04	0.54	0.04	0.24	0.02
Hidden precursor	0.55	0.41	0.45	0.51	0.52	0.00	0.53	0.05	0.45	0.04	0.24	0.03
Hidden pre	0.54	0.41	0.45	0.52	0.52	0.06	0.53	0.10	0.40	0.04	0.24	0.03
Future-state R ²												
H-shape R ²												
Mechanism F1												
ΔU R ²												

c

Future-state closure gain



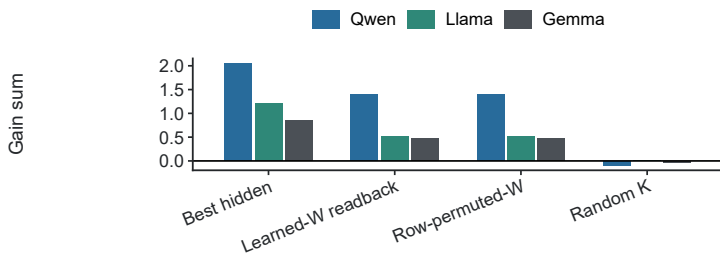
d Mechanism-label gain



e

Secondary composite summary

Sum of four metric-specific gains



f

Bounded result

Hidden-state proxies add more closure than readbacks

Global target PCA; other audits remain open